

ANSHUL KAUSHAL

AI ENTHUSIAST

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EDUCATION

Panjab University

B.E. Computer Science & Engineering

2022 - 2026

CGPA: 8.7 / 10

EXPERIENCE

IISER Bhopal

Project Intern: Deep Learning for Medical Imaging

Jun 2024 - Jul 2024

Bhopal, Madhya Pradesh

Guide: Prof. Vinod K Kurmi

- Worked on the Mamba state-space model for a comprehensive multi-domain medical imaging dataset.
- Enhanced conventional CNN-Transformer-based image segmentation models by implementing adaptive, advanced learnable fuzzy systems.
- Achieved substantial accuracy improvements while effectively reducing FLOPs, memory consumption, and the overall parameter count for better computational efficiency.

The Sparks Foundation

Data Science & Business Analytics Intern

Mar 2024 - Apr 2024

Remote

- Conducted textual sentiment analysis utilizing the Hugging Face RoBERTa model, alongside numerical prediction employing an LSTM model
- Implemented a vertical approach by integrating both methodologies and fine-tuning hyper-parameters to enhance accuracy
- Attained an R-squared metric exceeding 0.95, denoting a strong correlation between predicted and actual values

PUBLICATIONS

- "Optimizing Emotion Recognition in EEG Data: A Genetic Algorithm Approach with XAI Insights," The 15th International IEEE Conference on Computing, Communication and Networking Technologies (ICCCNT)

PROJECTS

COVID-19 Lung Infection Segmentation

- Developed various convolutional neural network architectures, including U-Net variants such as Trans U-Net, Residual U-Net, U-Net Squared, and Recurrent U-Net, without relying on pre-existing models
- Attained a high level of accuracy (Jaccard Index), surpassing 0.87, demonstrating the effectiveness of the implemented architectures in accurately segmenting the target features on COVID-19 Lung Dataset

Rock-Paper-Scissor Classifier

- Explored t-SNE (t-distributed stochastic neighbor embedding) plots, a dimensionality reduction technique, and Grad-Cam (Gradient-weighted Class Activation Mapping) visualization to gain insights into the learned features of the Rock, Paper, Scissors classifier
- Leveraged a pre-trained ResNet-18 neural network for feature extraction, benefiting from the model's learned representations
- Implemented data augmentation to modify the dataset enhancing the model's ability to generalize and improve its performance on Rock, Paper, Scissors classification tasks
- Successfully achieved an accuracy of 85% on training dataset and good real-world generalization of the model

RAG Chatbot

- It seamlessly integrates OpenAI's GPT-3.5 Turbo API for advanced language processing. In addition, used OpenAI's text embeddings to serve as a robust foundation to store and retrieve data
- Using Pinecone Vector Database for storing embedded information facilitates adaptive learning and precise information retrieval through dot product comparisons, ensuring the chatbot consistently delivers highly relevant responses

Spam-Ham Detector

- Acquired knowledge of Long Short-Term Memory (LSTM) networks, a type of recurrent neural network (RNN), and successfully implemented this model
- Explored and applied essential text data preprocessing techniques, including lemmatization to reduce words to their base form, tokenization for breaking text into individual words or tokens, removal of symbols, and eliminating common stop words

COURSES & CERTIFICATIONS

DeepLearning.AI (Coursera): Unsupervised Learning, Recommenders, Reinforcement Learning

Imperial College London (Coursera): Mathematics for machine learning: Linear Algebra

Google (Coursera): Crash Course on Python

Amazon Web Services: AWS Elements of data science

Duke University (Coursera): Data Science Math Skills

SKILLS

Technical Skills

- **Frameworks:** PyTorch, TensorFlow, Keras, OpenCV, Manim (Basics), Hugging Face
- **Libraries:** NumPy, Pandas, Seaborn, Matplotlib, Scikit-learn, Pygame, Tkinter
- **Programming Languages:** Python, C, C++, HTML, CSS, JavaScript
- **Tools:** MySQL, YOLOv8, Excel

Soft Skills

- **Communication and leadership skills:** Training and Placement cell member & ex. Model United Nation (MUN) committee-X organizer